



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

ASSIGNMENT NO 3- FALL 2025

Course Title:	Operating System			Course Code:	CSC323	Credit Hours:	3(2,1)
Course	Engr. Ahmad Mudassir			Program	BCE		
Semester:	5 th	Batch:	FA23-BCE	Section:	A,B	Date:	3 rd Jan 2026
Deadline	5 th Jan 2026			Maximum Marks:	20		
Student's Name:				Reg. No.	CUI/	/LHR	

Important Instructions / Guidelines:

Note: Assignments must reflect each student's individual work. Any instance of plagiarism, duplication, or copying from other students or online sources will result in a zero (0) grade for all parties involved.

Assignment Title: Threads Synchronization

Description

Analyze and compare different thread synchronization mechanisms used in operating systems, including **mutex locks, semaphores, and monitors**.

Your response should **differentiate** their conceptual design, **examine** their strengths and limitations, and **evaluate** their suitability for preventing race conditions in modern multi-core systems.

Finally, **justify** which synchronization mechanism is most appropriate for large-scale, concurrent software systems, based on correctness, maintainability, and scalability considerations.

Key Concepts Tested

- Thread creation and synchronization (pthread_create, pthread_join, pthread_mutex_t)
- Directory traversal (opendir(), readdir(), closedir())
- File I/O (read(), write())
- Interprocess communication (optional pipes)
- Resource sharing and protection (mutex/semaphore)

Deliverables

1. Source code files (.c)
2. Report including objective, working principle, flowchart, and output screenshots.

Test Case Table

Test Case ID	Input Directory	Description / Condition	Expected Output	Evaluation Criteria
TC1 – Basic	/testdir1	Directory with 5 files and 1 subdirectory.	Total files = 5, Total directories = 1, Total size matches actual.	Counts match; output format correct.
TC2 – Nested Recursion	/testdir_nested	Nested directories with multiple levels.	All nested files counted correctly.	Recursive traversal works; no missing files.
TC3 – Concurrency Stress	/testdir_concurrent	5 large subdirectories processed by threads.	Threads execute concurrently; total time < sequential.	Parallel execution verified; no data corruption.
TC4 – Empty / Invalid	/emptydir or /nonexistent	Empty directory or invalid path.	Prints “No files found” or error message.	Graceful handling; no crash.
TC5 – Performance	/testdir_large	Directory with 1000+ files.	Correct output within 2s (8 threads).	Improved performance; correct synchronization.

Supporting Test Setup Scripts

Use the following Bash commands to create test directories to run test cases

```
mkdir -p testdir1/sub
```

```
touch testdir1/file{1..5}.txt
```

```
touch testdir1/sub/file6.txt
```

```
mkdir -p testdir_nested/sub1/sub2/sub3
```

```
touch testdir_nested/sub1/file1.txt testdir_nested/sub1/sub2/file2.txt
```

```
mkdir -p testdir_concurrent/{A,B,C,D,E}
```

```
for d in testdir_concurrent/*; do touch $d/file{1..20}.txt; done
```

```
mkdir testdir_large
```

```
for i in {1..1000}; do echo "Hello" > testdir_large/file$i.txt; done
```

```
mkdir emptydir
```

Evaluation Criteria (Rubric)

Component	Marks	Description
Thread creation and synchronization	15	Correct usage of pthreads, mutexes
Directory traversal logic	10	Correct use of opendir/readdir
Report & Output correctness	10	Clarity, correctness, readability
Total	35	